

Human Capital Disclosure and Earnings Response: Evidence from SEC Regulation S-K

Final Project Presentation

Yeyang Zhou

Econ 5253: Data Science for Economists
University of Oklahoma

May 5, 2026

Research question. Do firms with greater *pre-mandate* human capital disclosure (HCD) experience a larger increase in their earnings response coefficient (ERC) after the SEC's 2020 mandate?

ERC — a quick gloss.

Conceptually: how strongly the stock market revises its valuation when a firm announces earnings. High ERC $\Rightarrow$ investors treat earnings as an informative, credible signal of firm value; low ERC $\Rightarrow$ earnings move prices little (noisy, or already known).

Operationally: the slope β in $\text{Return} = \alpha + \beta \cdot \text{Earnings surprise} + \varepsilon$.

Background.

- ▶ Pre-2020: SEC required only employee *headcount* in 10-Ks (checklist regime).
- ▶ Nov 2020 — Reg S-K Item 101(c) shifts to a *principles-based* regime: firms must describe “human capital resources” material to their business.
- ▶ No specific metrics prescribed $\Rightarrow$ wide cross-sectional variation in disclosure depth across firms.

Disclosure Examples: Same Mandate, Different Responses

Excerpts from FY2021–2025 10-K Item 1, “Human Capital” section.

Low Disclosure

Intercontinental Exchange (FY2025)

$hc_intensity = 0.009$

“...we had a total of 12,844 employees... less than 1% is subject to collective bargaining agreements, and **such relations are considered to be good...**”

Mid Disclosure

HEICO Corp. (FY2025)

$hc_intensity = 0.017$

“...approximately 11,100 employees... **Health and Safety:** striving for zero employee injuries... **Compensation and Benefits:** market competitive total rewards programs...”

High Disclosure

Trane Technologies (FY2021)

$hc_intensity = 0.041$

“**25.5% of global employees were women; 36.4% U.S. racially/ethnically diverse...** key talent retention **94.6%...** **89% participated** in engagement survey...”

Same mandate, very different disclosures $\Rightarrow$ **cross-sectional variation in HCD_i** , the source of identification.

Sources

- ▶ **SEC EDGAR**: 10-K filings, FY2015–2024 (text)
- ▶ **Compustat**: firm financials (earnings, market value)
- ▶ **CRSP**: monthly returns (for buy-and-hold abnormal returns)

Pipeline (Python)

- 1 Pull 10-K filings via the EDGAR REST API (rate-limited, locally cached).
- 2 Extract Item 1 (“Business”) with regex + iXBRL cleaning.
- 3 Tokenize; greedy phrase-match against the Demers, Wang & Wu (2024) lexicon (1,285 HC keywords across 5 categories).
- 4 Compute filing-level intensity: $hc_intensity_{i,t} = \frac{\# \text{HC keywords in Item 1}}{\# \text{words in Item 1}}$.
- 5 Merge with Compustat and CRSP at the firm-year level.

Final sample: $\sim 33,000$ firm-year observations, $\sim 4,000$ unique firms.

Treatment exposure (predetermined, FY2015–2019, standardized):

$$HCD_i = \frac{1}{5} \sum_{t=2015}^{2019} hc_intensity_{i,t}$$

Triple-interaction regression:

$$\begin{aligned} AR_{i,t} = & \alpha + \beta_1 UE_{i,t} + \beta_2 Post_t \times UE_{i,t} + \beta_3 HCD_i \times UE_{i,t} \\ & + \beta_4 Post_t \times HCD_i + \beta_5 \mathbf{Post}_t \times \mathbf{HCD}_i \times \mathbf{UE}_{i,t} \\ & + \mu_i + \lambda_t + \varepsilon_{i,t} \end{aligned}$$

- ▶ $AR_{i,t}$: 9-month buy-and-hold market-adjusted return (3 months after FYE)
- ▶ $UE_{i,t}$: earnings (IB) scaled by lagged market value of equity
- ▶ $Post_t = \mathbb{I}[t \geq 2021]$
- ▶ Firm FE μ_i + year FE λ_t ; SE clustered by firm; estimated via `linearmodels.PanelOLS`

Prediction (H1): $\hat{\beta}_5 > 0$ — firms with deeper pre-mandate HCD experience a larger ERC increase after the mandate.

	Main		Excl. FY2020	
	± 3 yrs	± 5 yrs	± 3 yrs	± 5 yrs
β_1 : <i>UE</i>	-0.020	0.039*	-0.069**	0.009
β_2 : <i>Post</i> $\times$ <i>UE</i>	-0.104***	-0.127***	0.081**	-0.016
β_3 : <i>HCD</i> $\times$ <i>UE</i>	-0.048	-0.061**	-0.033	-0.054**
β_4 : <i>Post</i> $\times$ <i>HCD</i>	-0.042***	-0.032***	-0.029***	-0.023***
β_5 : <i>Post</i> $\times$ <i>HCD</i> $\times$ <i>UE</i>	0.090**	0.103***	0.036	0.071**
<i>N</i>	17,554	25,392	14,998	22,836

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. SE clustered by firm.

- $\hat{\beta}_5 > 0$ and significant in main specification (both windows).
- Result **robust** to excluding FY2020 in the wider ± 5 -year window.

What did we learn? Why does it matter?

Empirical finding

- ▶ Firms with deeper voluntary HCD *before* the mandate experience a differentially larger ERC increase *after* it.

Why it matters

- ▶ **Disclosure regulation design:** principles-based mandates are *not* neutral — their value relevance depends on firms' prior disclosure infrastructure.
- ▶ **Capital markets:** voluntary disclosure becomes more informative once formalized through mandatory channels (legal liability, audit).
- ▶ **Policy:** informs ongoing debate around the SEC's 2022 proposal to expand mandatory human-capital metrics.

Limitations & next steps

- ▶ Add firm-level controls (size, leverage, analyst coverage, IO).
- ▶ Run an event-study / parallel-trends specification.

Thank you!

Comments and questions welcome.